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ROLE OF CORPORATE GREEN FINANCING IN ACHIEVING SUSTAINABILITY GOALS

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ABSTRACT:

This paper examines how green financing helps companies achieve their sustainability goals from economic and environmental angles. Businesses are under increasing pressure to be "green," and financial tools like green bonds and sustainability loans have become key ways to fund renewable energy, energy-efficient technology, and cleaner operations. However, implementing green finance is not always easy. High setup costs, complex regulations, and the risk of "greenwashing" (where companies only appear eco-friendly) often create obstacles. By analyzing financial reports and real-world cases, this research shows how companies can use green funds to stay profitable while improving their environmental impact. The study finds that firms transparent about their carbon footprint tend to receive better financing terms. At the same time, small businesses often struggle because the paperwork and technical requirements for green loans are demanding, leaving only larger firms able to access these benefits. The research also examines investor behavior, showing that investors now look for "future-proof" companies rather than just short-term profits. However, widespread skepticism due to greenwashing remains a challenge. Without a standard government framework or "green guidebook," companies may find it difficult to report their sustainability efforts honestly, making it hard for genuinely responsible firms to stand out. In conclusion, green finance represents a fundamental shift in corporate strategy and the global economy, not just a trend. Inclusive financing that enables smaller companies to participate is crucial. By bridging high-level policy with everyday business operations, green financing can move from being a luxury for large corporations to standard practice for all firms. The cost of doing nothing is far higher than the investment needed for green technology, making it essential for long-term business survival and environmental protection.

Keywords: *Green Financing, ESG Standing, Greenwashing Risks, Sustainable Growth, Environmental Regulations.*

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1. Introduction:

In the modern corporate world, the pressure to adopt environmentally sustainable practices has evolved from a voluntary choice into a strategic necessity. Global frameworks such as the United Nations Sustainable Development Goals (SDGs) (United Nations, 2015) and the Paris Agreement (UNFCCC, 2015) have established stringent climate targets, compelling businesses to balance profitability with environmental responsibility. However, for most companies, the primary challenge is not a lack of intent but the significant capital required to upgrade existing technologies and infrastructure. In this context, corporate green financing has emerged as a crucial mechanism, utilizing instruments such as green bonds and sustainability-linked loans to mobilize capital for environmentally sustainable transitions (Climate Bonds Initiative, 2023).

In India, the urgency of adopting green finance is even more pronounced. The country has committed to achieving net-zero emissions by 2070 (Government of India, 2021), a goal that requires an estimated cumulative investment exceeding USD 3 trillion (NITI Aayog, 2022). While major metropolitan areas have been the primary focus of green finance adoption, regional industrial hubs face more complex challenges. In these regions, the economy is often heavily dependent on transportation networks and industries such as steel and coal, which are traditionally high-emission sectors. While new policies are beginning to emphasize eco-friendly packaging and fuel-efficient transport systems, local firms continue to face constraints such as high initial investment costs and the technical complexities associated with implementing green technologies.

Despite the clear advantages of green finance, its implementation is often constrained by several structural challenges. One of the most significant issues is greenwashing, where companies misrepresent their environmental performance to gain financial or reputational benefits (OECD, 2022). Additionally, regulatory ambiguity and limited transparency in the allocation and utilization of green funds contribute to uncertainty among stakeholders. These factors create a "trust gap" between investors and businesses, thereby slowing the adoption of green financial instruments in emerging industrial regions.

This research aims to address this gap by examining how corporate green financing can support industries in achieving their environmental objectives. By analyzing the practical trade-off between economic growth and sustainability, the study seeks to develop a framework for the effective use of green financial instruments. Furthermore, it highlights how improved access to green finance can enhance Environmental, Social, and Governance (ESG) performance. Ultimately, the study demonstrates that with appropriate financial mechanisms and policy support, even traditionally high-emission industrial sectors can transition toward a more sustainable and economically viable future.

2.Objectives:

- To evaluate the growth of green bonds and sustainability-linked loans in India and see how these financial tools are helping the country reach its national climate goals.
- To find out the main reasons why companies are struggling to adopt green finance, specifically looking at high startup costs and the confusion caused by a lack of clear government regulations.

- To investigate how "greenwashing" and a lack of clear information affect investor trust, and whether this makes corporate sustainability reports less effective.
- To review the impact of RBI and SEBI guidelines on creating a more transparent system for green investments and supporting long-term sustainable growth.

3. Literature Review and Research Gap

Sustainable corporate development has become a global environmental necessity, as the world faces a significant "climate funding gap" where trillions of dollars are required to transition away from carbon-intensive economies (United Nations, 2023; World Bank, 2024). In the Indian context, this challenge is particularly acute; the nation requires an estimated cumulative investment of over USD 3 trillion by 2030 to meet its nationally determined contributions (NDCs) and a total of USD 10.1 trillion to reach Net-Zero by 2070 (NITI Aayog, 2022; CEEW, 2023). To mitigate these financial risks, regulatory frameworks such as the SEBI (Listing Obligations and Disclosure Requirements) and RBI's Framework for Acceptance of Green Deposits (2023) have established strict standards for how companies must report and utilize sustainable funds. In today's time, there is increasing pressure on companies to follow sustainable practices, and this has made green finance more important than before. It is no longer only about environmental responsibility but also about long-term business survival and growth. Modern financial solutions address these sustainability goals through integrated instruments including: Green Bonds: Fixed-income instruments specifically earmarked to raise money for climate and environmental projects. Sustainability-Linked Loans (SLLs): Lending instruments that incentivize the borrower's achievement of ambitious, predetermined sustainability performance targets (SPTs).

ESG Integration: The systematic inclusion of Environmental, Social, and Governance factors in financial analysis to improve long-term risk-adjusted returns (OECD, 2022). These financial instruments are helping companies move towards more sustainable operations in a practical way. For example, green bonds are commonly used to finance renewable energy projects like solar panels and wind energy. Sustainability-linked loans help companies stay motivated to achieve their environmental targets by linking them with financial benefits. ESG integration also helps investors make better decisions by looking at long-term risks instead of only short-term profits.

Corporate green financing generally follows a strategic sequence designed to transition firms toward low-carbon operations:

Capital Mobilization: Focuses on securing low-cost debt through green-labeled instruments to fund renewable energy and waste management.

Technological Integration: Employs green funds to upgrade "hard-to-abate" sectors (like steel and cement) with energy-efficient machinery and carbon-capture technologies.

Monitoring and Reporting: The final phase involves transparent disclosure of environmental outcomes, often leveraging digital tracking and third-party audits to ensure funds are used as promised (Climate Bonds Initiative, 2023).

This process shows how companies gradually shift from traditional methods to more sustainable practices. First, they arrange funds, then they invest in better technology, and finally they report their environmental performance. This step-by-step approach makes the transition more structured and easier to manage.

Green financing provides a better alternative to traditional commercial debt, particularly for funding "deep-green" innovations that have high initial costs. The primary advantage of these instruments lies in their ability to lower the Weighted Average Cost of Capital (WACC) for firms that demonstrate high environmental performance. Furthermore, the rise of Sustainability-Linked Bonds (SLBs) allows for flexibility, as the interest rates are tied directly to the company's ability to meet specific ESG targets, which lowers long-term operational costs for responsible firms (Flammer, 2021; Yang et al., 2025). From a corporate perspective, green financing is not only a financial tool but also a way for companies to plan their long-term growth while taking care of the environment. It helps them improve both their financial position and their public image.

Despite their potential, several gaps hinder the widespread adoption of green financing in regional industrial hubs:

The Risk of Greenwashing: There is a lack of standardized data and "taxonomy" (definitions), leading to cases where firms misrepresent their environmental impact to secure cheaper funding (OECD, 2022).

Regulatory Ambiguity: Most policies are designed at a national level; there is a need for more regional frameworks to help small and medium-sized industries in hubs like Raipur navigate complex compliance.

High Compliance Costs: The administrative burden of tracking and reporting green outcomes often prevents smaller firms from accessing the green bond market, leaving them reliant on expensive traditional debt.

These challenges make it difficult for many companies, especially smaller ones, to adopt green financing. Many firms do not have enough resources or knowledge to meet the requirements of green finance. At the same time, unclear rules create confusion, which reduces the confidence of both companies and investors. This shows that while green finance has many benefits, there are still practical problems that need to be solved.

The Research Gap: While there is a growing amount of research on green finance in India, most studies focus mainly on general concepts like green bonds, ESG reporting, and sustainability policies. There is limited research that clearly connects these financial tools with actual firm-level outcomes, especially across both environmental performance and financial benefits together. Most existing studies also focus on large corporations and national-level trends, and they do not fully cover how green financing affects small and medium enterprises or firms in regional industrial areas. As a result, the real challenges faced by smaller firms in accessing green finance are not properly highlighted. Another gap is that many studies discuss issues like greenwashing and regulatory frameworks, but there is less detailed analysis on how these problems directly affect investor trust and financing costs in real market situations. Our study helps to fill this gap by focusing on how green financing instruments such as green bonds and sustainability-linked loans are actually being used by companies in India, and how they impact ESG performance, financing cost, and transparency in practice.

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4. Research Problem:

Traditional corporate financing models often struggle to support long-term sustainability goals due to high interest rates and a focus on short-term profits. While green bonds and sustainability-linked loans (SLLs) are marketed as modern solutions to bridge the climate funding gap, they are often inaccessible to regional firms due to high setup costs and complex reporting requirements. Furthermore, the rising risk of "greenwashing"—where environmental claims are exaggerated—undermines investor confidence and prevents capital from reaching genuine eco-friendly projects. There is a critical gap in establishing a transparent, standardized, and cost-effective framework that prevents financial misrepresentation while ensuring that green capital is effectively utilized by corporations to meet their ESG targets and national climate commitments.

5. Methodology:

i. Data Identification and Sourcing: The study begins by identifying authoritative financial and regulatory documents. Key secondary sources include official reports published by the Reserve Bank of India (RBI), SEBI, and the Ministry of Finance. These sources are selected because they provide reliable and up-to-date information, helping establish a clear baseline for the growth of green financing in India between 2020 and 2025.

ii. Choosing Corporate Reports: To understand how green funds are utilized in practice, the research examines Business Responsibility and Sustainability Reports (BRSR) along with annual financial statements from major Indian industrial sectors. This step is important as it enables the tracking of green bond proceeds and evaluates how effectively these funds are allocated toward sustainability-related projects.

iii. Analytical Framework: The study adopts a mixed-method approach to ensure a comprehensive analysis.

- Trend Analysis: A quantitative assessment of the rise in green bond issuances and sustainability-linked loans (SLLs) to measure market expansion over time.
- Content Analysis: A qualitative review of corporate disclosures to identify recurring challenges, such as regulatory ambiguity, lack of standardization, and high initial investment costs.
- ESG Benchmarking: A comparative evaluation of Environmental, Social, and Governance (ESG) scores to examine whether access to green financing translates into measurable sustainability performance.

iv. Assessment of Barriers: To ensure accuracy and credibility, the study cross-references corporate environmental claims with third-party ESG audit reports. This process helps identify instances of greenwashing and highlights the existing trust gap between investors and corporations in emerging markets.

v. Synthesis and Framework Development: In the final stage, the findings are synthesized to develop a practical framework aimed at improving the effectiveness of green financing in India. The proposed framework includes policy recommendations focused on enhancing transparency, reducing regulatory uncertainty, and making green finance instruments more accessible to industries pursuing long-term sustainable growth. Expected Outcomes

6. Expected Outcomes:

- Financial Landscape Mapping: The study is expected to provide a clear understanding of the growth trends of green bonds and sustainability-linked loans (SLLs) in India over the past five years.
  - Barrier Identification:

It will identify key financial and structural challenges, such as high initial investment costs and regulatory uncertainty, that limit the adoption of green financing by firms.

- Mitigation of Greenwashing:

The research aims to highlight how improved ESG reporting standards and third-party audits can help reduce greenwashing and strengthen investor confidence.

- Scalable Framework: A practical and cost-effective framework is expected to be developed, enabling businesses to better utilize green finance for achieving sustainability goals.
- Impact on Performance: The findings are likely to demonstrate a positive relationship between the use of green funds and improvements ESG performance.

7.Data analysis and Findings

7.1 Growth of Green Bonds in India

Based on the trend analysis of SEBI and RBI reports, green bonds have moved from a niche concept to a major financial tool. In the last few years, India has become one of the largest emerging markets for green bonds. For example, the Government of India's issuance of Sovereign Green Bonds in 2023 showed that there is a high demand for these instruments.

India issued approximately ₹16,000 crore in Sovereign Green Bonds in 2023 compared to around ₹4,000 crore green bond issuances in earlier years (around 2020), showing a growth of approximately:

$(16000 - 4000) / 4000 \times 100 = 300\%$

However, the data shows that most of this growth is coming from the power and energy sectors. Large corporations use these bonds because they can handle the high cost of "Green Certification." Small businesses (SMEs) are almost completely missing from this data because they cannot afford the extra paperwork required to prove a project is "green."

Finding 1: Green bonds are growing fast in India, but the market is currently dominated by large energy companies, leaving smaller industries behind.

7.2 ESG Performance and Cost of Financing

By comparing corporate financial reports with their ESG (Environmental, Social, and Governance) scores, a clear pattern emerges. Companies that invest in transparency—meaning they clearly report their carbon emissions—often enjoy a "Greenium" This is a "green premium" where they get lower interest rates compared to traditional loans.

Companies with high ESG scores may receive loans at approximately 6.5% interest, while lower ESG-rated firms may pay around 8.5%, showing a difference of: $8.5\% - 6.5\% = 2\%$ lower cost of capital

Our analysis of recent bank lending trends shows that banks are now more willing to give better terms to "future-proof" companies. However, this only works if the company has a high ESG score. If a company has a low score or refuses to disclose its data, investors see them as a high risk, leading to higher interest costs.

Finding 2: Higher ESG transparency is directly linked to lower financing costs, proving that being sustainable is financially profitable for Indian firms.

7.3 Comparison: Green Finance vs. Traditional Financing

When we compare traditional loans to green loans, the main difference is the "payback period" and the "intent." Traditional financing is easier to get but focuses only on short-term profit. Green finance, like Sustainability-Linked Loans (SLLs), offers lower rates but requires the company to meet specific environmental targets (like reducing water waste by 20%).

Over a 5-year period, a 2% lower interest rate on a ₹100 crore loan can save:

$100 \times 0.02 \times 5 =$ ₹10 crore in interest savings

Data shows that while green finance is cheaper in the long run, the "entry cost" is higher. Traditional loans don't require expensive third-party audits, making them the default choice for companies that are in a hurry or have limited budgets.

Finding: Green finance is strategically better for long-term growth, but traditional financing is still more practical for daily business needs due to less regulation.

7.4 Challenges: Greenwashing and Regulatory Ambiguity

One of the biggest problems found in this analysis is the "Trust Gap." By cross-checking corporate claims with third-party reports, we found instances of "Greenwashing." Some companies use green funds for general corporate purposes rather than specific eco-friendly projects.

Global ESG studies suggest that around 20% of sustainability disclosures show signs of incomplete or misleading reporting, indicating a reliability issue in sustainability claims. This happens because India currently lacks a single, standardized "Green Guidebook" or Taxonomy. While the RBI has released frameworks, there is still confusion among businesses about what exactly counts as a "green" activity. This confusion increases the risk for investors and makes them skeptical of corporate sustainability reports.

Summary of Findings

The analysis of this study leads to several important conclusions. First, green financing has become an important tool for companies to achieve long-term sustainability while also maintaining their financial performance. The growth of green bonds in India shows that there is increasing demand for environmentally responsible investments, especially in sectors like power and energy. Second, the study finds that companies with higher ESG transparency are able to benefit from better financing terms. Firms that clearly report their environmental performance are seen as more reliable by investors and banks, which helps them get loans at lower interest rates. This shows that sustainability is not only an environmental concern but also a financial advantage. Third, the comparison between green finance and traditional financing shows that green finance is more beneficial in the long run, even though it has higher initial costs. Traditional loans are still more commonly used because they are easier to access and require less compliance.

Another important finding is that small and medium enterprises face major challenges in accessing green finance. Due to high certification costs, complex procedures, and lack of technical knowledge, most SMEs are unable to take advantage of green financial instruments. As a result, large corporations continue to dominate this space. Finally, the study highlights the issue of greenwashing and lack of clear regulations. The absence of a standardized framework creates confusion among companies and reduces investor trust. This makes it difficult to clearly identify which companies are genuinely sustainable. Overall, the findings suggest that while green financing has strong potential to support sustainable growth in India, its success depends on improving transparency, reducing regulatory confusion, and making these financial tools more accessible to all types of businesses.

8. Limitations of the Study

Every research study has certain limitations that need to be acknowledged to understand the scope and reliability of the findings. In this study, a few important limitations were observed.

The research is based mainly on secondary data collected from reports published by institutions such as RBI, SEBI, and other government sources. While these sources are reliable, they do not always provide detailed or company-specific data, especially for small and medium enterprises (SMEs). This makes it difficult to fully analyze how green financing impacts smaller firms. There is a limitation in terms of data availability and standardization. Since India does not yet have a single unified framework or clear taxonomy for green finance, the information reported by different companies may not be fully consistent or comparable. This creates challenges in accurately evaluating ESG performance across firms. Another important limitation is the difficulty in identifying and measuring greenwashing. Although the study attempts to cross-check corporate claims with available reports, it is not always possible to verify whether companies are truly using green funds for sustainable purposes. This lack of complete transparency can affect the accuracy of the analysis. The study focuses on a limited time period (2020–2025), which may not fully capture the long-term impact of green financing. Since green investments often take time to show results, the conclusions drawn may not reflect future outcomes. Finally, due to time and resource constraints, the research does not include primary data such as surveys or interviews with industry experts. Including such data could have provided deeper insights into the practical challenges faced by companies in adopting green finance.

9. Conclusion and Future Scope

9.1 conclusion:

The study concludes that green financing plays an important role in supporting sustainable development in India. It helps companies to raise funds for environmentally friendly projects while also improving their financial performance. The increasing use of green bonds and sustainability-linked loans shows that businesses are slowly moving towards more responsible and sustainable financial practices.

The study also shows that companies with better ESG transparency are able to get better financing conditions, which proves that sustainability is linked with financial benefits. At the same time, green finance is still not equally accessible to all firms. Large companies are more able to use these instruments, while small and medium enterprises face difficulties due to high costs and complex procedures.

Another important conclusion is that green finance is more effective in the long term compared to traditional financing, even though it requires higher initial investment and compliance. However, issues like greenwashing and lack of clear regulations still reduce trust in the system and affect its full effectiveness. These challenges also slow down the overall adoption of green financial instruments in developing markets like India.

Overall, green financing is not just a financial tool but a necessary step towards achieving long-term environmental and economic balance in India. Its success depends on better transparency, stronger regulations, and wider access for all types of businesses.

9.2 Future Scope

While green financing has established a strong foundation in India, the journey toward a fully sustainable economy is still in its early stages. The "Future Scope" of this research identifies several critical areas where the financial ecosystem must evolve to reach India's 2070 Net-Zero targets.

1. Transition from Secondary to Primary Research:

This study relied primarily on secondary data from RBI and SEBI reports. There is significant scope for future researchers to conduct primary research, involving deep-dive interviews and surveys with Chief Financial Officers (CFOs) and sustainability heads. This would provide "on-the-ground" insights into the practical trade-offs companies face when choosing between a green bond and a traditional loan, moving beyond what is simply stated in annual reports.

2. Longitudinal Impact of the SEBI 2025 Framework:

Since the SEBI "ESG Debt Securities Framework" was only recently implemented in 2025, its long-term impact on market transparency is not yet fully visible. Future studies should monitor the next five years of corporate filings to determine if these new rules actually reduce greenwashing. Research should track whether the "Trust Gap" between investors and companies closes as a result of these stricter disclosure requirements or if firms find new ways to bypass regulations.

3. Sector-Specific "Greeniums" and Performance:

This research focused on the general corporate landscape, but there is a need for sector-specific analysis. Future research could compare high-emission sectors like Steel and Cement against the Renewable Energy sector to see if the "Greenium" (interest rate discount) is consistent. Analyzing whether a "green-certified" steel plant actually performs better financially than a traditional one over a 10-year period would provide the empirical proof needed to convince more conservative investors.

4. Financial Inclusion for the SME Sector:

A major finding of this paper was that Small and Medium Enterprises (SMEs) are currently left out of the green finance boom. Future scope exists in developing and testing a "Scalable Green Framework" specifically for smaller firms. Researchers should investigate whether "Group Green Bonds"—where multiple small firms borrow money together—could lower the high entry and certification costs that currently act as a barrier for individual small businesses. In summary, green finance in India has moved past the stage of being a "trend" and is now a strategic necessity. However, for management and policymakers, the focus must now shift from simply issuing bonds to ensuring the integrity of the funds used. Future studies will play a vital role in perfecting the tools needed to ensure that India's economic growth remains both inclusive and environmentally sustainable.

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